

DIGITAL TWIN CHALLENGE 2026

ChipUP

A Digital Twin for Predictive Glucose Spike Monitoring
in Type 2 Diabetes

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Glucose spikes are treated after the fact

Type 2 Diabetes affects over 100 million people in India. One of its most dangerous everyday risks is the post-meal glucose spike — a rapid rise that silently damages the heart, kidneys, and eyes over time.

Today, patients and doctors usually find out about a spike only after it has already happened — via a CGM reading or a lab test. By then, the damage is already underway.

100M+

People living with Type 2 Diabetes in India

Reactive

is how care works today — react after the spike, not before it

Our goal: proactive

predict the spike 2 hours before it happens

What is a patient Digital Twin?

A Digital Twin is a continuously updated virtual replica of a patient's physiology — built by fusing two very different kinds of data into one living model.

Static / Historical

The patient's long-term clinical profile

- Demographics & BMI
- HbA1c & years since diagnosis
- Medication (insulin, metformin)
- Comorbidities (BP, cardiovascular)
- Lipid panel & kidney markers

+

Dynamic / Real-time

The patient's live physiological trajectory

- Continuous glucose (CGM), 5-min
- Heart rate & HRV
- Sleep stage
- Step count / activity
- Updated continuously, not once

Fused into one continuously updated patient twin

ChipUP: predicting the spike, not just the sugar

We built a proof-of-concept Digital Twin focused on one specific, high-impact prediction task: will this patient's glucose spike in the next 2 hours?

1

Simulate the twin

Generate synthetic EHR + wearable data mimicking real patients

2

Fuse the streams

Merge rolling wearable windows with each patient's static profile

3

Predict the event

XGBoost model flags spike risk 2 hours ahead of onset

4

Inform the doctor

Risk score + explainability surfaced on a live dashboard

Navigating the sandbox rules

Real patient data is restricted by privacy laws like the DPDP Act and HIPAA. Every dataset in this project is synthetically generated — no real patient information is used anywhere.



For EHR data

Static clinical profile generation

- Synthetic patient generation (Synthea-style)
- 200 simulated patient profiles
- Realistic HbA1c, BMI, medication mix



For wearable data

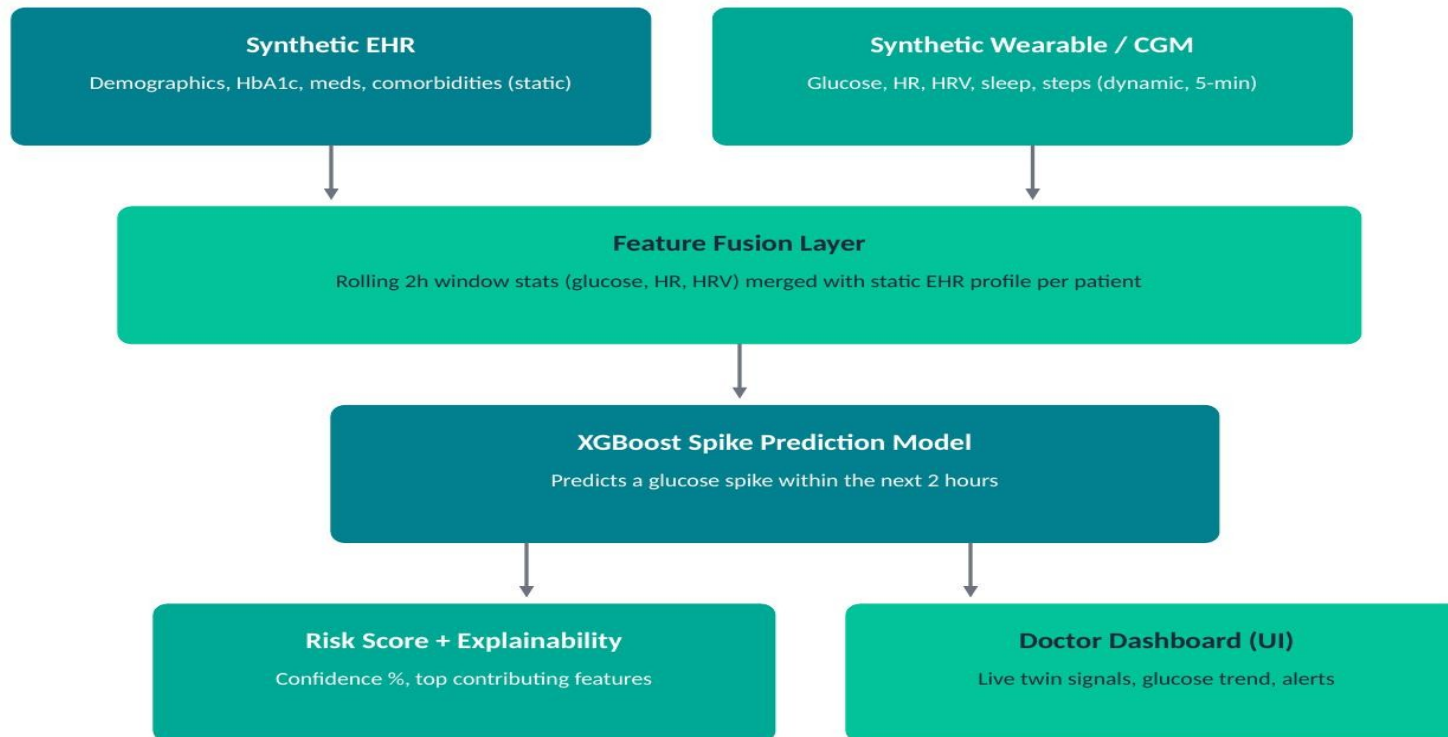
Dynamic time-series generation

- 5-minute interval CGM + HR + HRV + sleep
- 7 days per patient (~403K readings)
- Spike frequency scaled to HbA1c severity

End-to-end architecture

ChipUP Digital Twin — System Architecture

Type 2 Diabetes glucose-spike prediction pipeline



Data Sandbox Compliance

All EHR and wearable data is synthetically generated. No real patient records are used at any stage, in line with the DPDP Act and HIPAA sandbox rules for this challenge.

Continuous digital twin loop

New wearable readings arrive every 5 minutes, re-triggering fusion and re-scoring the twin — risk stays continuously current, not a one-time snapshot.

Why XGBoost for this prediction task



Fast iteration

Trains in seconds on CPU — critical for a hackathon timeline



Explainable

Feature importances build clinician trust, unlike a black box



Imbalance-aware

scale_pos_weight handles the natural rarity of real spikes



Strong tabular fit

Rolling-window features already encode short-term temporal signal

Technical stack

Language	Python 3
Data generation	NumPy, Pandas
Feature engineering	Rolling-window stats (Pandas)
Model	XGBoost (gradient-boosted trees)
Evaluation	scikit-learn (ROC-AUC, PR-AUC)
Dashboard mockup	HTML / CSS

Model performance on held-out patients



Top predictive features



Known limitation (by design)

Precision on the rare spike class is currently low at the default threshold, due to natural class imbalance. Next steps: threshold tuning, cost-sensitive learning, and a larger synthetic cohort. We surface this openly rather than hide it.

How a doctor interacts with the twin



From PoC to real-world impact

Clinical impact

- Pre-emptive dietary or insulin guidance
- Fewer emergency interventions
- Better long-term glycemic control
- Builds clinician trust via explainability

Near-term roadmap

- Validate on real, IRB-approved cohorts
- Threshold tuning for higher precision
- Expand to hypoglycemia prediction
- Multi-day trend forecasting

Long-term vision

- Extend to hypertension, CV risk twins
- Full-body multi-condition twin
- Clinician co-design of the dashboard
- Integration with real EHR systems

Thank you

ChipUP — Digital Twin Challenge 2026

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